

Frank Shan

✉ g3shan@uwaterloo.ca ☎ +1-263-881-6666 in frank-shan 📄 knarfthekant 🌐 portfolio

HIGHLIGHTS

- Engineer focused on extensible deep learning and applied AI systems, including retrieval, neural modeling, and developer tooling. Builds practical products that connect ML workflows with reliable backend systems and measurable delivery outcomes.

EDUCATION

University of Waterloo | *CGPA: 92.3/100* *Sep 2024 - Apr 2029*
Bachelor of Computer Science, Honours Computer Science, AI Specialization

EXPERIENCE

Software Developer Intern | *Richmond, BC* *Sep 2025 - Apr 2026*
Ziyutec Inc.

- Built a **RAG agent**, indexing **500+ enterprise documents** of different format; improved **contextual relevance by 16%** (DeepEval) via **hybrid search and reranking** on synthetically generated evaluation set.
- Built an automation agent for engineering workflow, reducing operational workload by **20%** across a **50-person** engineering team.

Research Assistant | *Waterloo, ON* *Feb 2026 - Present*
University of Waterloo

- Designed comparative experiments for a probabilistic programming language PAL that compiles structural rules into tensor operations. Built a **hybrid neuro-symbolic** framework for training Named Entity Recognition models to enforce strict linguistic constraints and improve early-epoch model convergence.
- Structured reproducible experiments and analysis pipelines for model comparisons, enabling faster iteration on constraint-driven NER training strategies.

PROJECTS

Resume Generator v1 | *Python, Jinja2, prompt_toolkit, LaTeX, LangGraph* [github](#) 

- Added an AI-ready selection stage so the system can match resume content to job descriptions before rendering and compilation; adapted this workflow to support a vLLM-based project integration.

SKILLS

Programming Languages: `built-in method items of dict object at 0x109c32c00`

AI and ML: `built-in method items of dict object at 0x109cec980`

Frameworks: `built-in method items of dict object at 0x109c32c80`

Platforms: `built-in method items of dict object at 0x109e3c0c0`

PROFESSIONAL CERTIFICATES

Stanford Online *December 2025*
XCS224N Natural Language Understanding [credential](#) 